

Date:21/04/2026

Submitted by: Arya Sharma

Mobile Forensic

1. Introduction

Mobile forensics is a subdomain of digital forensics, which concerns the identification, preservation, acquisition, and analysis of mobile data. Mobile devices include smartphones, tablets, and SIM cards, which hold significant amounts of personal data (call logs, location, multimedia, and application data). Mobile forensics without forensic tools refers to conducting forensic investigation manually. This practice is highly applicable in cases when there are no forensic tools available or restrictions apply to their use. Moreover, this type of investigation is useful for conducting preliminary analysis in case when further investigation requires the use of forensic software.

2. Types of Mobile Data

In manual mobile forensics, the emphasis is put on determining various types of data present in the device:

Communication Data: Phone calls, SMS, MMS, email, and other instant message apps (WhatsApp, Telegram).

Multimedia Data: Pictures, videos, and audio files.

Applications Data: Social networking applications, banking applications, browsing history.

System Data: Information about the device itself (IMEI number, software version, list of applications installed).

Location Data: GPS trails, mapping information, geotagging of pictures.**3.**

3. Methodology

1. Identification and Seizure

- Determine device and take it off networks (Airplane mode).
- Prevent any remote wiping or tampering.
- Note device status (powered-on/powered-off, locked/unlocked).

2. Preservation

- Minimize interactions with the device.

- Employ Faraday bags (where available) or remove connectivity.
- Keep chain of custody record.

3. Documentation

- Document the following:
- Model of device, IMEI number
- Information about SIM card
- Notifications and screen view
- Photograph the device from different views.

4. Manual Acquisition

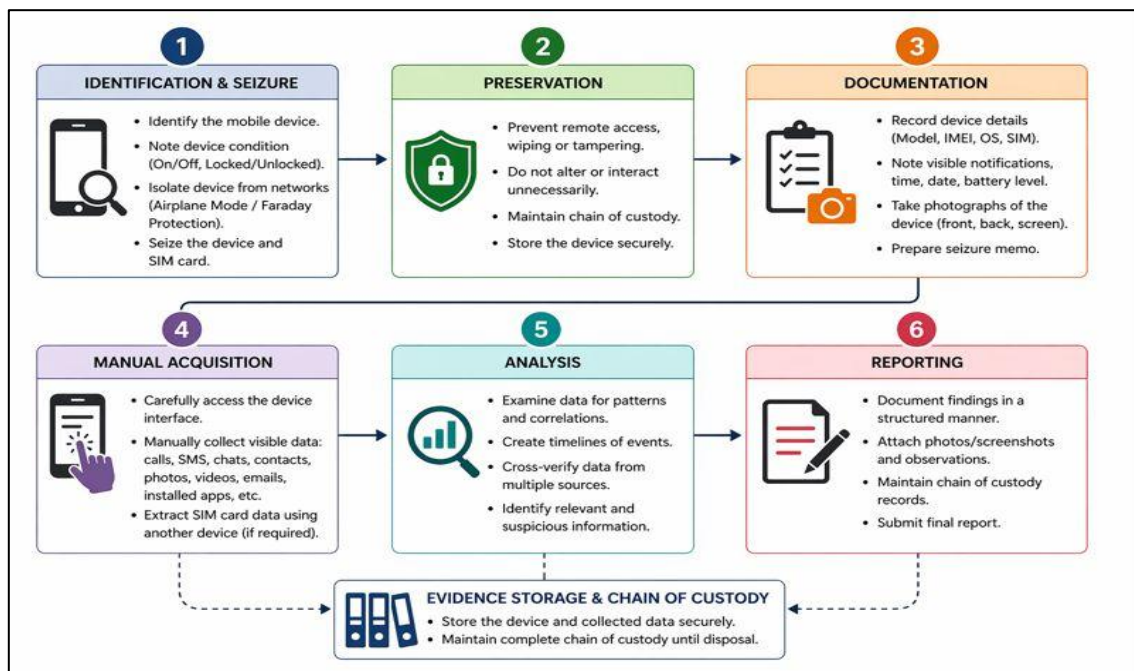
- Navigate device interface carefully.
- Acquire visible information including:
- Call history
- Messages
- Gallery
- Contact list
- Browsing history

5. Analysis

- Seek commonality among data:
- Regularly contacted persons
- Unusual or alarming messages
- Chronology of incidents
- Cross-reference findings manually.

6. Report Writing

- Provide a comprehensive report that includes:
- Observations
- Photographs/snapshots
- Chronology of events



4. Techniques

1. Visual Checkup

- Monitor notifications on lock screen.
- Review recent app launches and running processes.

2. Manual Browsing

- Browse through apps manually without any data change.
- Look for:
 - WhatsApp messages
 - SMS
 - Phone Call Logs

3. Metadata's Analysis

- Monitor timestamps, file size, and location of files.
- Find trends from communication.

4. SIM Card Analysis

- Remove SIM card and analyze through other devices.
- Access to contacts and messages stored on SIM.

5. Cross Verification

- Cross-verify information obtained from various apps such as SMS vs WhatsApp.



- Verify timestamp consistency.

6. Isolate Device from Network.

5. Applications

➤ Cyber Crime Investigation

Analysis of messages, fraudulent transactions, and malicious applications.

➤ Harassment/Threatening Case

Collection of messages sent through chats and phone calls.

➤ Corporate Investigation

Investigating data leaks and other corporate insider crimes.

➤ Missing People

Tracking last contact and geographical location.

➤ On-Field Investigation

Real-time analysis before sending to the laboratory.

6. Challenges/Drawbacks

1. Possibility of Data Tampering

Human intervention can alter data timestamps or corrupt data.

2. Access Limitation

Locks prevent access.

Encrypted files cannot be viewed directly.

3. Lack of Deleted Data Recovery

Unable to recover data that has been erased without forensic tools.

4. User Errors

Misinterpretation of data.

Failure to capture crucial information.

5. Inefficient

Involves manual verification of all apps and documents.

6. Ethical Concerns

Wrong procedures can render evidence unusable in legal proceedings.

7. Benefits

- Not dependent on costly forensic software.
- Very useful for on-site investigations.
- Aids in preserving volatile data.
- Develops analytical and observational skills.

8. List of Mobile Forensic Tools

1. Commercial Tools

- Cellebrite UFED
- Cellebrite Physical Analyzer
- MSAB XRY
- Oxygen Forensic Detective
- Magnet AXIOM
- MOBILedit Forensic
- Elcomsoft iOS Forensic Toolkit
- GrayKey (iOS unlocking)

2. SIM Card & Storage Tools

- SIM Card Reader
- Dekart SIM Explorer
- Paraben SIM Card Seizure
- USB Memory Card Reader
- FTK Imager (for storage analysis)

3. Android-Specific Tools

- ADB (Android Debug Bridge)
- AFLogical OSE
- Andriller



- MobSF (Mobile Security Framework)

4. iOS-Specific Tools

- iTunes / Finder Backup
- Elcomsoft Phone Breaker
- GrayKey
- Cellebrite Premium (iOS support)

5. Open-Source Tools

- Autopsy
- Sleuth Kit
- Wireshark (network analysis)
- Volatility (memory analysis – indirect use)

6. Cloud & App Data Tools

- Oxygen Cloud Extractor
- Magnet AXIOM Cloud
- Belkasoft Evidence Center

7. Password & Unlocking Tools

- Cellebrite Advanced Services
- GrayKey
- Elcomsoft Toolkit

9. Lessons Learned

The manual approach to mobile forensics underscores some essential aspects of digital forensics, emphasizing discipline, precision, and observation as compared to reliance on sophisticated devices and software. First, it underlines the significance of evidence protection. As a matter of fact, even something seemingly innocuous, like turning off an application or connecting the device to the Internet, might change vital data like the time stamp and start a series of automatic updates. This means that a careful, step-by-step approach is required.

Second, the absence of automated logging of data, analysis and extraction results stresses the need for meticulous documentation. Indeed, unlike in other approaches, where the program

automatically logs the work performed, in manual forensics, the responsibility lies entirely on the investigator to document all observations made during the analysis. It implies that he or she needs to make detailed notes, take photos and preserve the chain of custody.

This approach will also enhance analytical skills and intuition in investigation. Since no sophisticated equipment can be used for filtering or emphasizing important information, the investigator will have to manually discover trends like repeated communication with a certain person or abnormal behavior during certain periods. This will increase the investigator's skill in linking the different pieces of information and constructing a logical narrative about what happened.

On the other hand, the weaknesses of manual forensics are also instructive. Inability to retrieve deleted files or decrypt any information suggests the necessity of using special tools in more complex cases. Moreover, manual forensics should be considered an introductory method or even a supplement to computer forensics, but not a full-scale approach.

In addition to this, it also instills the value of ethical behavior and professionalism in investigators. Dealing with one's personal phone entails handling sensitive data, and as such, it is important that confidentiality and integrity of the investigation process are maintained.

In summary, the toolless method of mobile forensics is one of the most basic methods of investigating through which essential investigative skills can be learned. Although the method is not perfect, it does hold a significant position in the field of mobile forensics.

Aviation Forensics

1. Introduction

Aviation forensics is a function area within the broader field of forensic science, specifically concerned with investigating incidents or accidents in aviation to ascertain their cause. It entails the scientific investigation of the physical evidence, the digital flight data, human factors, and the environment. The ultimate goal is not only determining the cause of an accident but also preventing any similar incident in the future.

Modern aircraft have sophisticated technologies that produce substantial amounts of data. For this reason, aviation forensics is data-driven and interdisciplinary, drawing from various fields such as aeronautical engineering, digital forensics, meteorology, and human psychology. Forensic investigations in aviation are normally carried out by aviation regulatory bodies like the National Transportation Safety Board and the Directorate General of Civil Aviation.

2. Types of Evidence in Aviation Forensic Investigations

- **Flight Data Recorder (FDR):** This type of evidence is used to document the data on speed, altitude, and engine performance of the aircraft.
- **Cockpit Voice Recorder (CVR):** It documents voice communications between the pilots and their conversations recorded within the cockpit.
- **Wreckage Evidence:** It documents structural damages to the wreckage, pattern of the debris, and impact marks.
- **Maintenance Evidence:** It documents any records related to the service history of the aircraft.
- **Air Traffic Control (ATC):** It documents communication between pilots and ATC.

3. Methodology

1. Securing the Location & First Steps

- Establish perimeter around crash location.
- Ensure safety (check for fuel and fires).

2. Gathering Evidence

- Remove black boxes (FDR & CVR).
- Obtain fragments and label them.
- Take photos to map the scene.

3. Recording Information

- Note how debris spread and angles.
- Control chain of evidence.

4. Extracting Information

- Obtain information from FDR & CVR.
- Get ATC logs and radar information.

5. Analysis

- Recreate the flight and time lines.
- Conduct technical analysis of failures & pilot maneuvers.
- Assess environmental factors.

6. Reconstruction

- Recreate physical aspects of airplane (actual or simulated).
- Conduct simulations based on gathered information.

7. Report Writing

- Create final report with results & conclusions.

4. Techniques Used

Investigation into Aircraft Crash and Analysis of Black Box

1. Black Box Analysis

- Analysing flight data and cockpit voice recording.
- Locating the anomalies prior to the aircraft crash.

2. Aircraft Debris Analysis

- Analysing fracture and burn patterns.
- Locating the site of impact and explosion.

3. Recreating Flight Trajectory

- Utilizing radar and flight data recorder information to recreate flight path.

4. Human Factors Analysis

- Evaluating human behavior like pilot decisions, stress, and fatigue.

5. Environmental Factors Analysis

- Evaluation of meteorological, wildlife, and turbulence factors.

6. Simulation Techniques

- Simulating aircraft accidents with computer modeling.

5. Use Cases

- **Aircraft Accident Investigation**

Finding out reasons for accidents (technical problems, pilot error).

- **Terrorism and Sabotage Cases**

Examining explosives and sabotage.

- **System Failure Investigations**

Finding out reasons for engine or other technical breakdowns.

- **Safety Improvements**

Improving air safety standards and airplane construction.

- **Legal and Insurance Cases**

Gathering necessary evidence for litigation.

6. Challenges & Limitations

1. Extensive Damage to Physical Evidence

Crashes typically damage essential parts.

2. Harder to Recover Information

Black boxes can be broken or under water.

3. Complicated, Multi-Disciplinary Interpretation

Needs cooperation among several specialists.

4. Effects of Weather and Environment

Temperature, water, and fire can affect evidence.

5. Long-Term Process

Extensive investigations can take years.



6. Fragile Nature of Information/Data

Information/data may never be recovered.

7. Advantages of Aviation Forensics

- Facilitates aviation safety
- Promotes prevention of accidents in future
- Causes accurate analysis
- Enables legal and regulatory measures
- Contributes to aircraft design and pilot training

8. Takeaways

The field of aviation forensics is among the most complicated fields in forensic science because it involves many disciplines. The first valuable point I have realized is how crucial systematic research is in this area. The analysis of each air accident is extremely accurate, including all kinds of physical proof, digital records, and human factor evidence. This means that it is extremely important to follow a methodology while conducting a forensic analysis.

One more important point from the reading is how significant the application of technology can be in investigating air accidents. Flight data recorder and cockpit voice recorder help researchers to learn what had really happened during the last moments of an accident. Nevertheless, it should be noted that to analyze this data, you should know everything about aircraft operation and its systems.

Finally, another thing I have learned is that human factor plays a very significant role in analyzing aviation accidents. Some air accidents do not result from mechanical failures but arise because of human mistakes, such as communication problems, wrong decisions, or excessive fatigue..

Forensic aviation also promotes preventive learning. Contrary to most other forensic disciplines, which concentrate their efforts on solving crimes, the investigation carried out within the framework of forensic aviation serves purposes of safety and the prevention of future aviation tragedies. Therefore, results of such studies usually lead to changes in aviation policies, improvements of pilot training courses, and enhancements of aircraft technologies. Thus, this forensic discipline can be called extremely useful for the benefit of society.

Another lesson learned from this field concerns evidence reconstruction. Aircraft accidents are frequently accompanied by large-scale destruction, implying that investigators have to assemble fragments of evidence carefully. Such process requires a lot of patience and skills of thorough analysis. The importance of saving even minute bits of evidence is highlighted here.

Lastly, aviation forensics teaches about the necessity for international cooperation, as the field itself is a global one, involving multiple stakeholders.

To conclude, forensic aviation implies lessons concerning preciseness, collaboration, and advancement of knowledge. This type of investigation goes beyond the determination of the cause of accidents, providing opportunities for future enhancement of aviation security.

Digital Electronic Meter Forensics

1. Introduction

The study of Digital Electronic Meter Forensics is an investigation technique used to analyze electric energy meters to check for any manipulation, fraud, and abnormal readings. Due to the advancement from analog meters to smart digital ones, utility companies are now able to gather a wide range of information about their metering systems such as logs and timestamps.

This field is particularly relevant for utility companies and law enforcements agencies as electricity theft could cause significant loss and misuse of electricity distribution between users. This particular field integrates knowledge of electrical engineering and computer forensics techniques.

2. Types of Evidence in Meter Forensics

- **Meter Reading Data** – Units consumed with respect to time
- **Event Logs** – Tampering incidents (cover opened, reversal of flow, magnetic field interference)
- **Load Survey Data** – Load profile per hour/day
- **Physical Evidence** – Malicious physical alterations
- **Firmware Data** – Meter firmware
- **External Connection** – Unauthorized bypass wiring

3. Methodology

1. Identification & Seizure

- Identification of suspicious meter based on abnormal meter readings
- Seizing and disconnection of meter
- Meter number, location, and consumer identification

2. Preservation

- Preventing tampering
- Maintaining chain of custody
- Safe storage for analysis

3. Documentation

- Photo documentation of meter status (seals, wires, case, etc.)
- Record readings and any damages
- Record environmental factors

4. Data Collection

- Readings manually taken
- Data from displays and logs (when available)
- Data obtained through meter itself

5. Analysis

- Comparison with past usage patterns
- Abnormalities in usage levels (drops or spikes)
- Tampering records

6. Correlation

- Bill and meter data of nearby consumers compared
- Loading behavior of customers correlated with meter reading

7. Reporting

- Drafting report to include:
 - o Evidences
 - o Tampering observations
 - o Graphs
 - o Conclu

4. Techniques Used

Inspection and Analysis Methods for Meters

1. Visual Examination

- Detection of broken seals, scratches, or other unauthorized access attempts
- Detection of bypassing or direct connection methods

2. Load Behavior Analysis

- Analysis of usage pattern changes
- Detection of abnormal consumption practices

3. Analysis of Event Logs

- Checking event logs for any tamper-related alerts
- Analysis of frequency and timing of events

4. Comparative Analysis

- Comparing consumption against:
 - o Other months
 - o Similar households/industries

5. Manual Reading Cross-Check

- Making sure that readings match consumption

6. Electrical Measurement

- Measuring current flow through the circuit

5. Uses of Meter Inspection Services

•Electrical Power Theft Detection

Detection of unauthorized power consumption via bypassing/tampering

•Billing Dispute Resolution

Settling disputes between consumers and service providers

•Industrial Consumption Monitoring

Checking conformity with the company's energy consumption policies

•Fraud Prevention

Detection of tampering with meters in commercial installations

•Regulatory Compliance Assistance



6. Challenges and Limitations

1. Advanced Manipulation Techniques

- Magnets or circuitry interference used for tampering

2. Insufficient Access to Internal Information

- Information inside some meters is not available

3. Environmental Factors

- Weather conditions, heat, and dust could corrupt information

4. Interpretation Difficulties

- Knowledge in electrical systems needed to analyze data

5. Legal Issues

- Evidences should be collected according to procedures

6. Technological Developments

- Smart meters call for advanced methods

7. Benefits of Meter Forensics

- Energy theft detection is efficient
- Cuts monetary loss
- Billing process is equitable
- Legal support provided
- Effective energy management

8. Takeaways

Meter Forensic Sciences is an area in the realm of forensic sciences where an exclusive look at the possibilities of its use outside the area of criminalistics is provided. One of the main lessons that meter forensics teaches is the power of data analysis in detection of crimes. The traditional crime scene analysis relies much on physical findings. In case with forensic meter examination, the investigator deals mainly with data analysis, which takes some special analytical skills.

One should note that in meter forensics, one needs to combine the analysis of physical and digital evidence. For example, there might be some physical indicators of fraud, such as broken seals or modified connections. However, in order to prove that the data obtained from the meter is fraudulent, one needs to analyze the logs and consumption data of the meter.

Finally, the lesson learned is about preventive monitoring. If meter readings are analyzed frequently enough, any anomalies will be found early and necessary measures will be taken before any losses are done.

It is notable that another important lesson learned from this topic is the influence of technological advancement. In recent years, contemporary electronic smart meters have become highly sophisticated, featuring such capabilities as remote reading, anti-tampering technology, and automatic data recording. On one hand, this improves forensic analysis by enabling easy access to crucial information; on the other hand, perpetrators will continue to become increasingly creative in their attempts to circumvent these technological innovations. As such, forensics experts need to stay abreast of the latest developments in technology and adjust their methods accordingly.

At the same time, it is important to recognize the legal and ethical issues involved in this area of study. Forensic evidence collected during the investigation of smart meters should be recorded correctly and preserved in accordance with established rules to ensure its admissibility in courts.

Another aspect of this topic worth considering is the societal contribution of forensic science. Indeed, the prevention of electricity theft helps ensure fair distribution of resources and alleviates the financial burden on legitimate consumers. Moreover, it contributes to the sustainable development of energy systems.

To conclude, digital electronic meter forensics exemplifies the practical application of forensic science principles to solve economic problems.